Venkata Krishna Chaitanya Nuthalapati

+1 404-490-5061 nvk.681@gmail.com linkedin.com/in/nvk-chaitanya github.com/nvk681

Technical Skills

Languages: Java, Python, Go, JavaScript/TypeScript, C++, SQL
Backend & Data: Spring Boot, Node.js, REST, gRPC, Kafka, RabbitMQ, PostgreSQL, MongoDB, Redis
Cloud & DevOps: AWS, Kubernetes, Docker, Jenkins, CI/CD, CloudFormation, Prometheus, Grafana

Experience

Workday **Aug 2025 – Present**
Software Development Engineer *Atlanta, GA*

- **CI/CD API Platform**: Developed production REST APIs supporting revision management, conflict detection, audit reporting, and controlled deployment workflows for an internal CI/CD platform.
- **Security & Reliability**: Strengthened service authentication and authorization, separated service and user access contexts, and added cancellation, error handling, and controlled recovery paths for deployment workflows.
- **Observability & Distributed Tracing**: Added monitoring metrics, dashboards, and alerts for service reliability; improved tracing across distributed pipeline components to support incident investigation.
- **Cloud & Async Reliability**: Optimized cloud service resource allocation, evaluated architecture migration considerations, and improved the reliability of asynchronous audit workflows through bounded execution, step-level job tracking, and failure handling.

Amazon **Sep 2023 – May 2025**
Software Development Engineer *Seattle, WA*

- **WRAP Migration**: Automated and optimized the **Lambda-based migration framework** with built-in validations for retail integrations in the UK, JP, and US, achieving an approximately **30% reduction in migration time**.
- **Guardrails**: Developed workflow rules and validations that automated error handling and data validation, resulting in approximately **60% fewer migration-related tickets**.
- **Grocery Listing**: Enhanced the listing process for the supply chain, cutting listing operation time from approximately **10 minutes to 4 minutes**.
- **Sourcing Grid**: Designed a new **API and user interface** that simplified sourcing configuration and attribute selection.
- **UNFI Enhancement**: Rebuilt UNFI review and deep dive pages on **AWS Lambda**, streamlining vendor item-store updates.
- **WRAP Backfills**: Led backfill initiatives in the UK, JP, and US, enabling online sales for physical stores.
- **Risks**: Proactively identified and resolved security and software issues across team-owned services.

RE IT Services Inc **Feb 2023 – July 2023**
Software Development Engineer *Atlanta, GA*

- **Data Pipeline**: Implemented auto archival of database records, resulting in an approximately **90% improvement in query performance**.
- **API Scaling**: Enhanced APIs to support increased load and automated tasks within the **CI/CD pipeline** using **Jenkins** and **Docker**, saving approximately **6 hours of manual work per task**.
- **Data Processing**: Developed **event-driven data processing** solutions using **Kafka** to standardize multi-source data across **18 systems**, reducing data discrepancies and cutting on-call workload by approximately **11 hours per week**.

Amazon **May 2022 – Aug 2022**
SDE Intern at AWS *Metro DC, VA*

- **AEA Onboarding**: Collaborated to streamline the **AWS Enterprise Access** onboarding process, reducing manual onboarding effort.
- **Topos**: Developed a full-stack console using **Python** to track and address security issues identified by network scans across internal services.

Darwinbox (Cloud-based HRMS platform) **Jun 2018 – Aug 2021**
Senior Software Engineer (promoted during tenure) *Hyderabad, India*

- **Microservices Migration**: Led the rebuild of the workflow engine as a **microservices architecture** using **Node.js** and **Docker**, shortening workflow execution time and enabling business workflows as a service.
- **Dynamic Reporting Framework**: Implemented a PHP-based reporting engine handling large on-demand data requests in a single operation.
- **Payslip Customization**: Developed a flexible framework for payslip customization to support large enterprise clients.
- **MongoDB Optimization**: Executed optimization strategies that reduced MongoDB usage and lowered infrastructure costs.
- **Multi-Payroll Integration**: Created **stateless APIs** for payroll systems, serving users across multiple payroll systems.
- **Payroll Automation**: Reduced manual payroll processing effort by automating payroll workflows using **Redis** and **RabbitMQ**.
- **Bulk Notifications**: Developed Bash scripts for bulk email and notification processing to support large-scale employee communications.

- **Taxation System**: Built a framework for handling multi-country tax systems, supporting the platform's expansion to clients in new countries.
- **Recruitment Automation**: Constructed a recruitment automation pipeline using **event emitters** to shorten the hiring cycle.

Darwinbox (Cloud-based HRMS platform) **Dec 2017 – Apr 2018**
Intern – Software Developer *Hyderabad, India*

- **Chatbot**: Leveraged **Wit.ai** (Facebook NLP tool) to build a chatbot that automated routine HR tasks, streamlining HR workflows using **Go**.
- **OCR Integration**: Integrated **OCR** into the reimbursement module, automating data extraction from bills and reducing manual processing.

Education

Georgia State University – Full Tuition Waiver and GTA **Aug 2021 – Dec 2022**
Master of Science in Computer Science; GPA: 3.96/4.0 *Atlanta, GA*

Vignan University – Merit Scholarship **Aug 2014 – Aug 2018**
Bachelor of Technology in Computer Science; Marks: 86.76% *Guntur, AP, India*

Personal Projects

Gumo | *Node.js, Elasticsearch, Neo4j, npm* **Apr 2021 – Present**

- Built and maintain an open-source (MIT) web crawler and scraper that extracts data from nested dynamic web pages for downstream knowledge-mining applications; first published on npm in Apr 2021, latest release v2.0.1 in Feb 2026.

NFL Health & Safety | *Python, Pandas, NumPy* **Dec 2021**

- Developed helmet segmentation, player tracking, and labeling methods for the NFL-AWS player safety challenge; findings later posted as an arXiv preprint.

Volunteering

Mentors Without Borders **Feb 2026 – Present**
Volunteer Instructor *Remote*

- Mentor junior high school students in Ghana through weekly remote sessions, teaching **web development and programming fundamentals** with hands-on exercises and structured projects.

Research Publications

- **Nuthalapati, V.K.C.** *Balancing Microservices and Monolithic Architectures.* IEEE Computer Society Santa Clara Valley Chapter, *Feedforward*, Vol. 5, No. 3, pp. 28–33, Jul 2026. Also available as arXiv:2607.03898
- **Mathur, M., Chandrashekhar, A.B., Nuthalapati, V.K.C.** *Real Time Multi-Object Detection for Helmet Safety.* arXiv preprint, May 2022. arXiv:2205.09878
- **Agrawal, A., Bhyravarapu, S., Nuthalapati, V.K.C.** (published as N.V.K. Chaitanya). *Matrix Hashing with Two Level of Collision Resolution.* 2018 8th International Conference on Cloud Computing, Data Science & Engineering (Confluence), IEEE, 2018. DOI: 10.1109/CONFLUENCE.2018.8442607
- **Vattikuti, B., Nuthalapati, V.K.C., Jukuntla, A.** (published as N.V.K. Chaitanya). *Installation Automation Tool.* Advances in Intelligent Systems and Computing, Springer, 2018. DOI: 10.1007/978-981-10-7868-2_24

Professional Service & Presentations

- **Peer Review**: Web of Science reviewer record: 26 reviews of 12 manuscripts (as of Sep 27, 2026); reviewer activity for IEEE Internet of Things Journal and JITCAI.
- **Technical Presentations**: MFPT 2026, *Resilient Data Infrastructure* (pre-recorded video, Jul 2026); AAC in the Cloud 2026, *Using AI Tools to Improve Accessibility and Communication Support* (live, Jun 2026).

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